

Biomechanical Reinforcement Learning

Simulating Movement-Based Interaction

S³CIX Summer School, Glasgow 2026

Florian Fischer, Arthur Fleig, Miroslav Bachinski

Welcome & Introduction



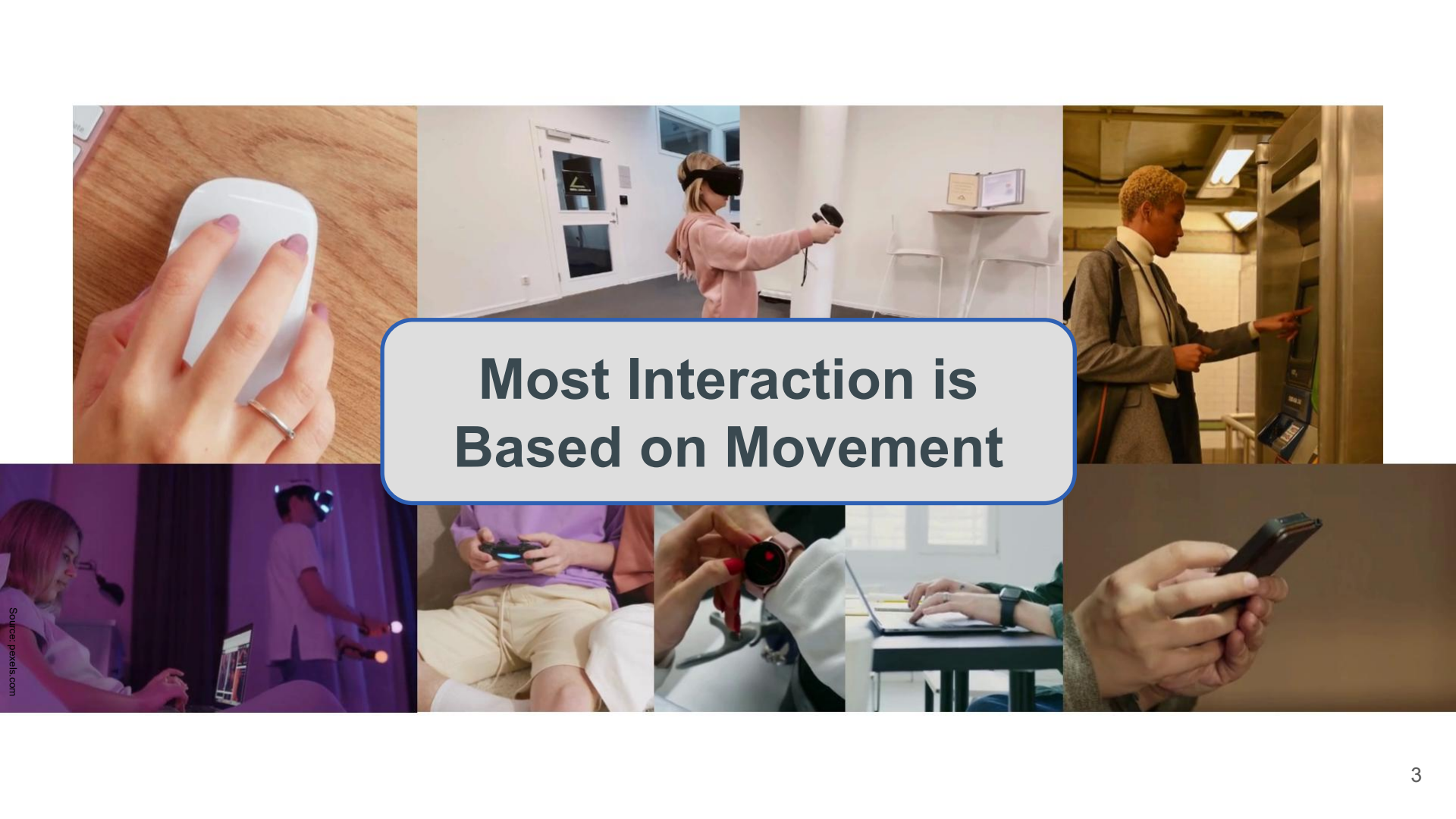
Florian Fischer
University of Cambridge



Arthur Fleig
ScaDS.AI Leipzig



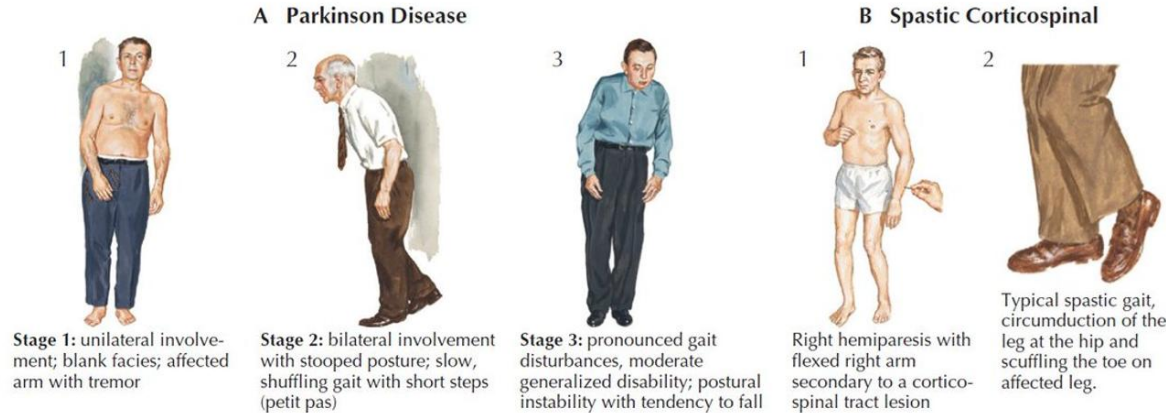
Miroslav Bachinski
University of Bergen



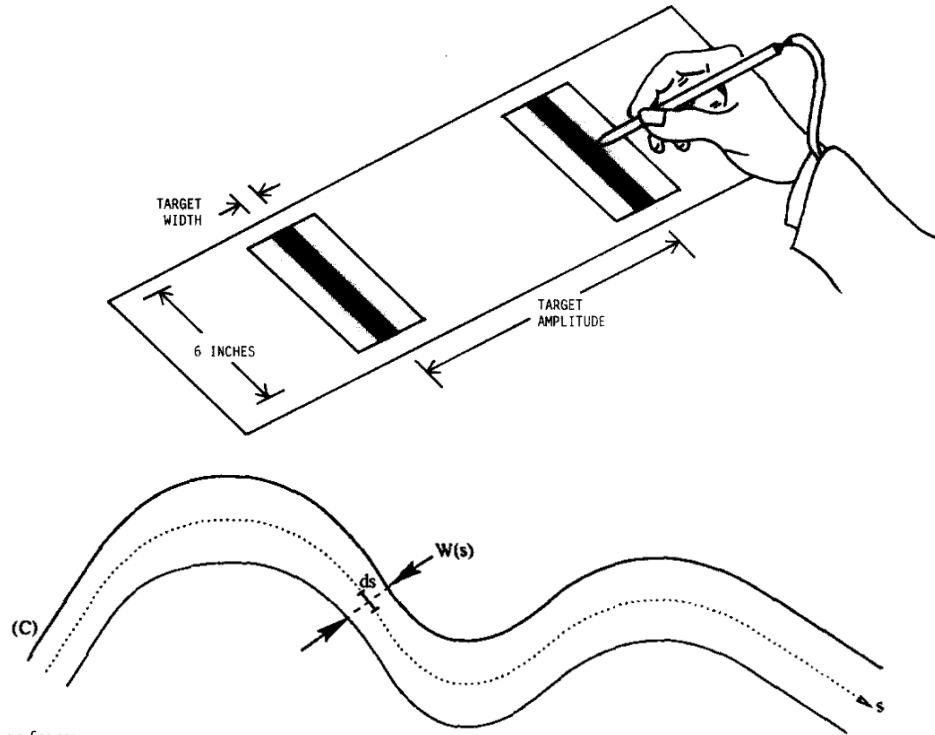
**Most Interaction is
Based on Movement**

Body Movement Data...

- **Conveys meaningful information** about user intent, uncertainty and well-being
- Can be used to **infer uncomfortable body postures and physical fatigue**



Traditional Understanding of (Movement-Based) Interaction



Pick up the mobile phone	<i>I</i>	1.18 sec.
Enter main menu	<i>M, H</i> [Hotkey]	1.35 sec. + 0.16 sec.
Go to 'Programs'	<i>M, 3H</i> [Hotkey]	1.35 sec. + 3*0.16 sec.
Select 'Programs'	<i>K</i> [Hotkey]	0.16 sec.
Go to 'Collection'	<i>M, H</i> [Hotkey]	1.35 sec. + 0.16 sec.
Select 'Collection'	<i>K</i> [Hotkey]	0.16 sec.
Select 'Choose program'	<i>K</i> [Hotkey]	0.16 sec.
...	...	...
Process tag	<i>R_{NFC}</i>	2.58 sec.
Find next tag	<i>M</i>	1.35 sec.
Movement to tag 'To'	<i>P</i>	1.00 sec.
NFC interaction	<i>A_{NFC}</i>	0.00 sec.
Process tag	<i>R_{NFC}</i>	2.58 sec.
Find next tag	<i>M</i>	1.35 sec.
...	...	...
Sum:		174.84 sec.

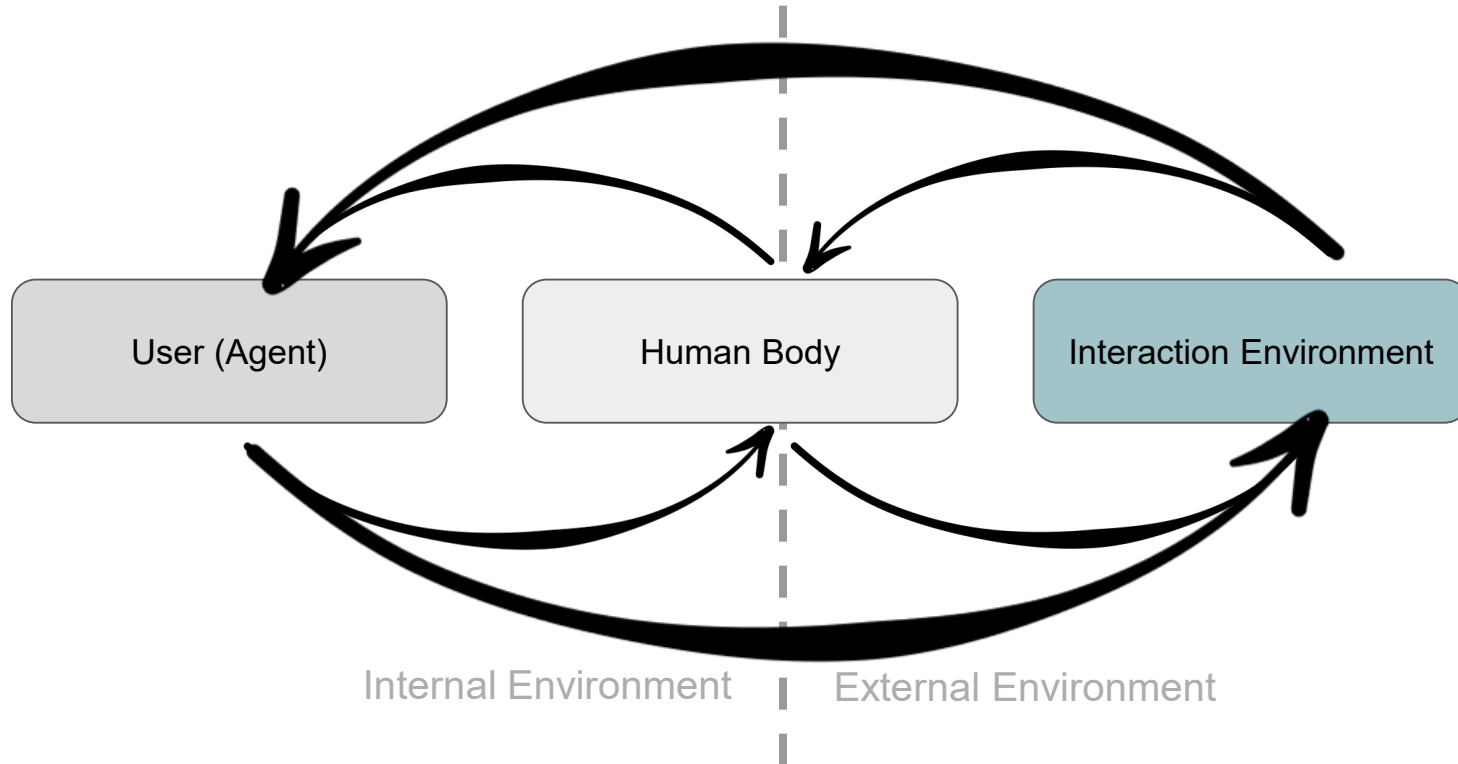
Images from:

Mackenzie, I. S. (1992). Fitts' law as a research and design tool in human-computer interaction.

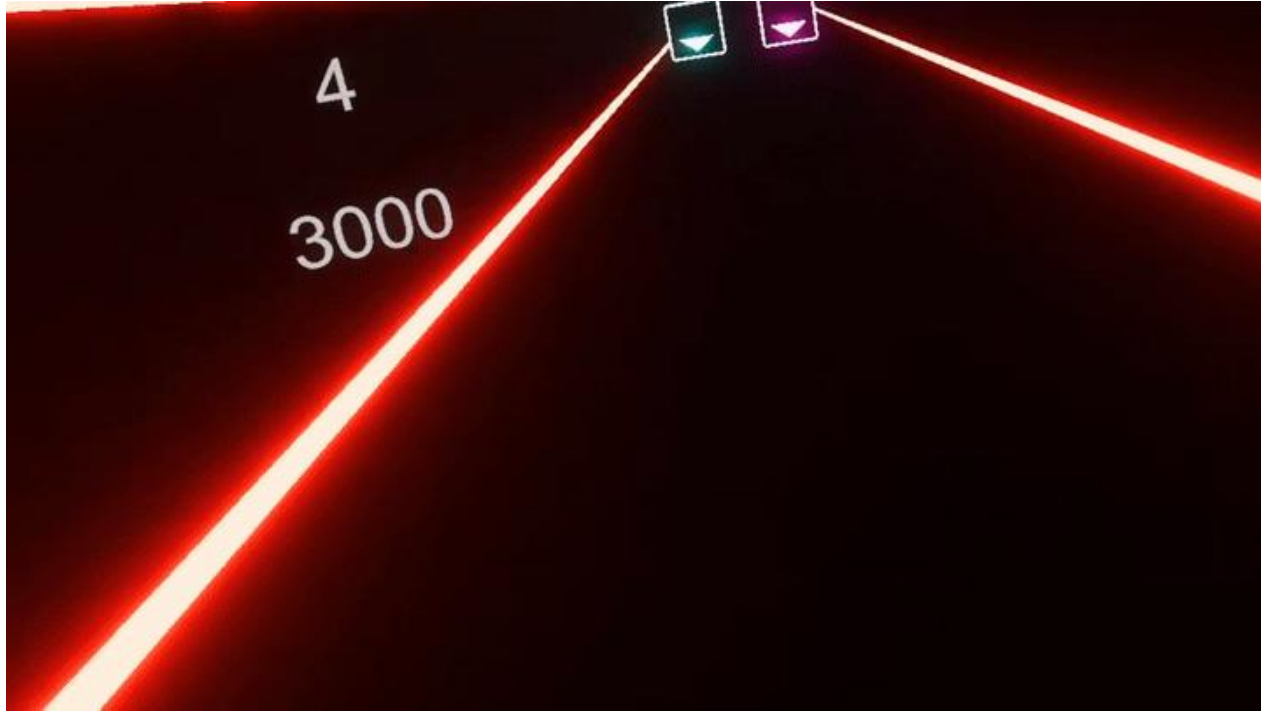
Accot, J., & Zhai, S. (1999). Performance evaluation of input devices in trajectory-based tasks: an application of the steering law.

Holleis, P., Otto, F., Hussmann, H., & Schmidt, A. (2007). Keystroke-level model for advanced mobile phone interaction.

HCI “with a body”



Example: Designing an Exergame



Biomechanical User Simulations in HCI

- We need to understand how humans use their body to interact with the environment
- Biomechanical models allow us to analyse existing interaction movements and predict performance, efforts, and strategies for a given design
- Content of this course:
 - What are biomechanical models?
 - How are they implemented in a physics engine?
 - How can we use them to predict how users interact with their environment?
 - How to infer flaws and limitations of a biomechanical interaction agent?

1. An Introduction to Biomechanical Models

Human Physiology

- Musculoskeletal system
- Passive tissues
- Neural control
- Energy delivery



Body segment

- Represents a segment between two joints, e.g., upper arm, lower arm, thigh, etc.
- Is characterized by:
 - Geometry
 - Mass
 - Mass center
 - Inertia matrix
- Body acceleration linked to the force/torque acting on it through the second Newton's law

$$\mathbf{F} = \mathbf{m} \mathbf{a} \text{ and } \boldsymbol{\tau} = \mathbf{I} \boldsymbol{\alpha}$$



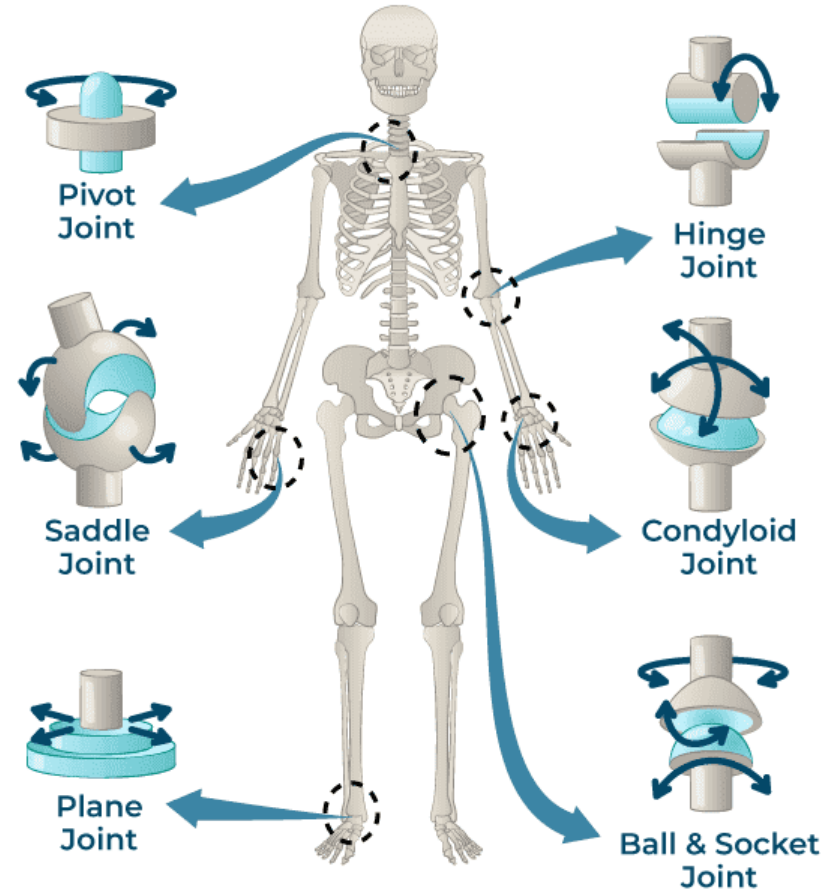
Skeletal system and joints

- Rigid bodies together compose a skeleton
- Connected through joints
- The core/pelvis is the root element
- Joints characterized by:
 - Semantic 1D-3D coordinate
 - Rigid transformation



Joint types

- Mechanical equivalents for allowed joint movement
- Define clear transformations between the reference frames
- Biomechanical models contain mechanical equivalents



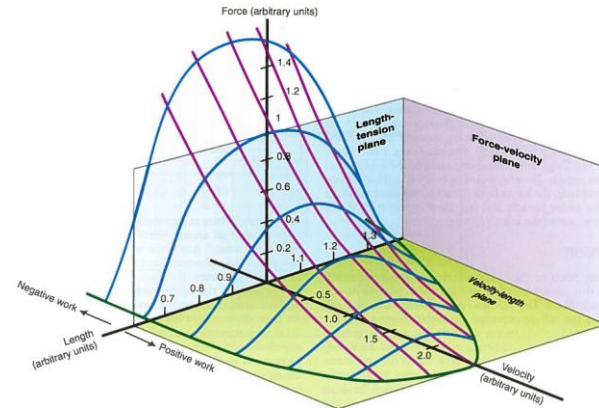
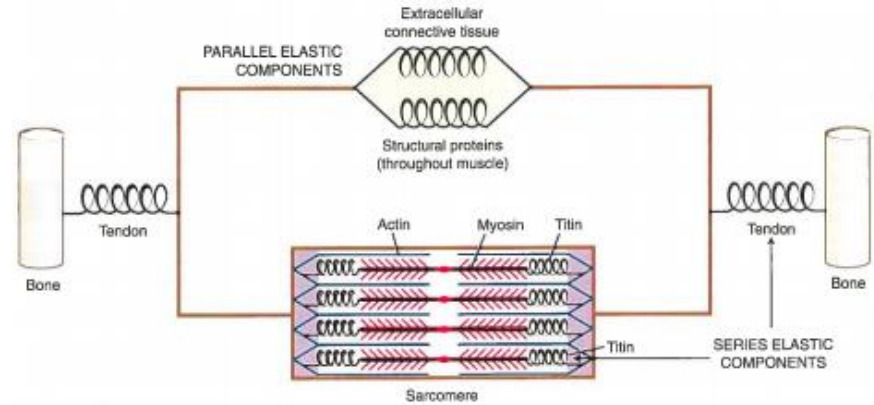
Muscle – stabilizer and mover of human body

- Muscles compete with external forces to keep the body static
- Muscles produce active forces for different limb movements
- Muscles with tendons provide elasticity for the human body
- Characterized by
 - insertion points, routing points, and wrapping geometry within the musculoskeletal system
 - Muscle shape is split into action lines



Hill-type muscle model

- A muscle is represented as:
 - Active contractile element (contractile proteins in muscle fibers)
 - Parallel elastic element (passive tissue between the muscle fibers)
 - Series elastic element (tendon)
- Characterized by:
 - Cross-sectional area – max isometric force
 - Optimal fiber length
 - Tendon slack length
 - Pennation angle at optimal fiber length
 - Tendon elasticity
 - Max contractile velocity
 - Passive elastic force curve



Ergonomics variables of interest

Joints:

- Extreme joint angles
- Instantaneous forces inside joints
- Instantaneous moments at joints
- Joint moments integrated over time
- Joint loads orthogonal to joint axis (lower row)

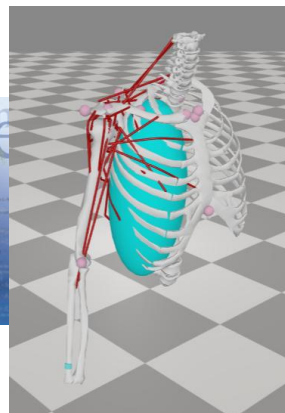
Muscles:

- Forces exerted by muscles
- Neural activations of muscles
- Total energy expenditure
- Muscular fatigue and recovery

State-of-the-art biomechanical simulators

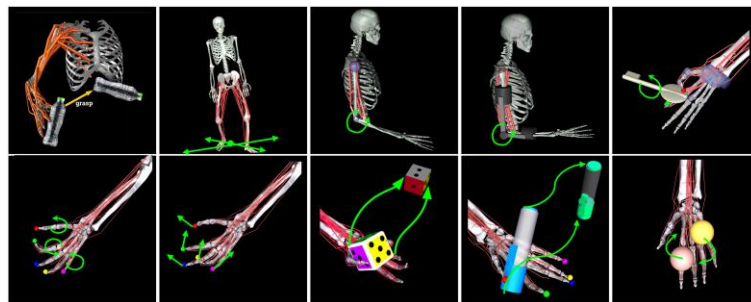
- OpenSim

- +Validated models
- +State-of-the-art fidelity
- +Model adaptation routines
- +Inverse simulation for recorded movements



- MuJoCo

- **+Fast for RL**
- +Good contact handling
- **+Interacting with environment**



Demo: MuJoCo Body Modelling

→ bit.ly/s3cixbio



2. Computational Methods for Forward Simulations

Human Controller

Internal Model



Dynamics f

Optimal Control Problem

$$\begin{aligned} \min_u J(x, u) \\ \text{s.t. } \dot{x} = f(x, u) \end{aligned}$$

Costs J

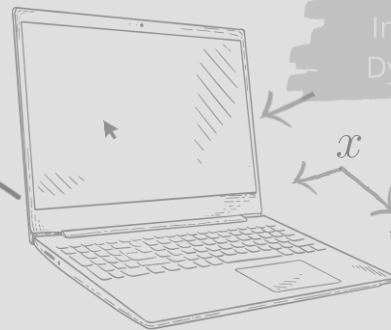
Perceptual Feedback y



Human-Computer System Dynamics f

Interface Dynamics

Input Device Dynamics



Muscle Control Signal u

Human Body Dynamics



Task

Human Controller

Internal Model

Interface Dynamics

Input Device Dynamics

Human Body Dynamics

Dynamics f

Control Strategy

$$u = \pi(x)$$

Perceptual Feedback y

Human-Computer System Dynamics f

Interface Dynamics

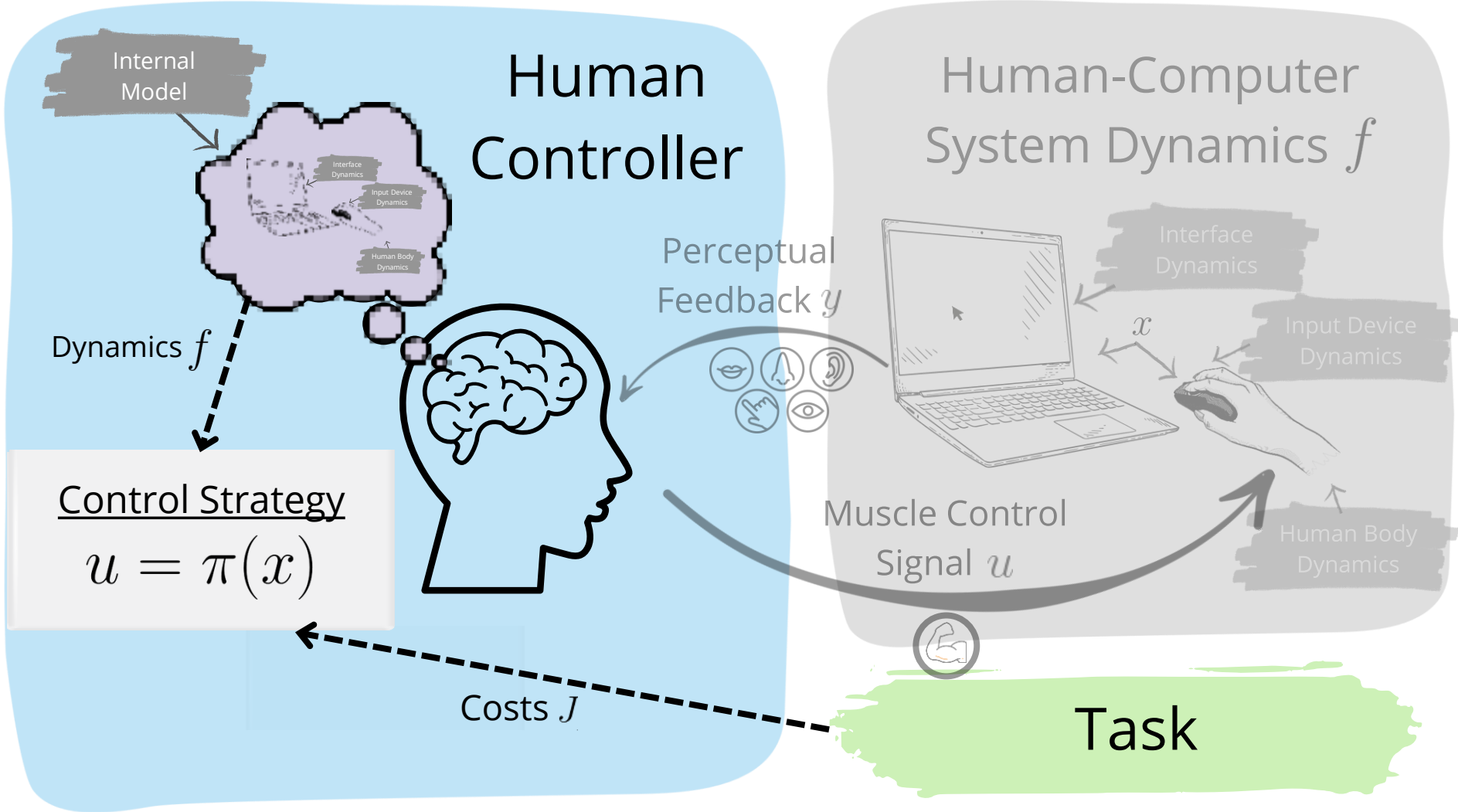
Input Device Dynamics

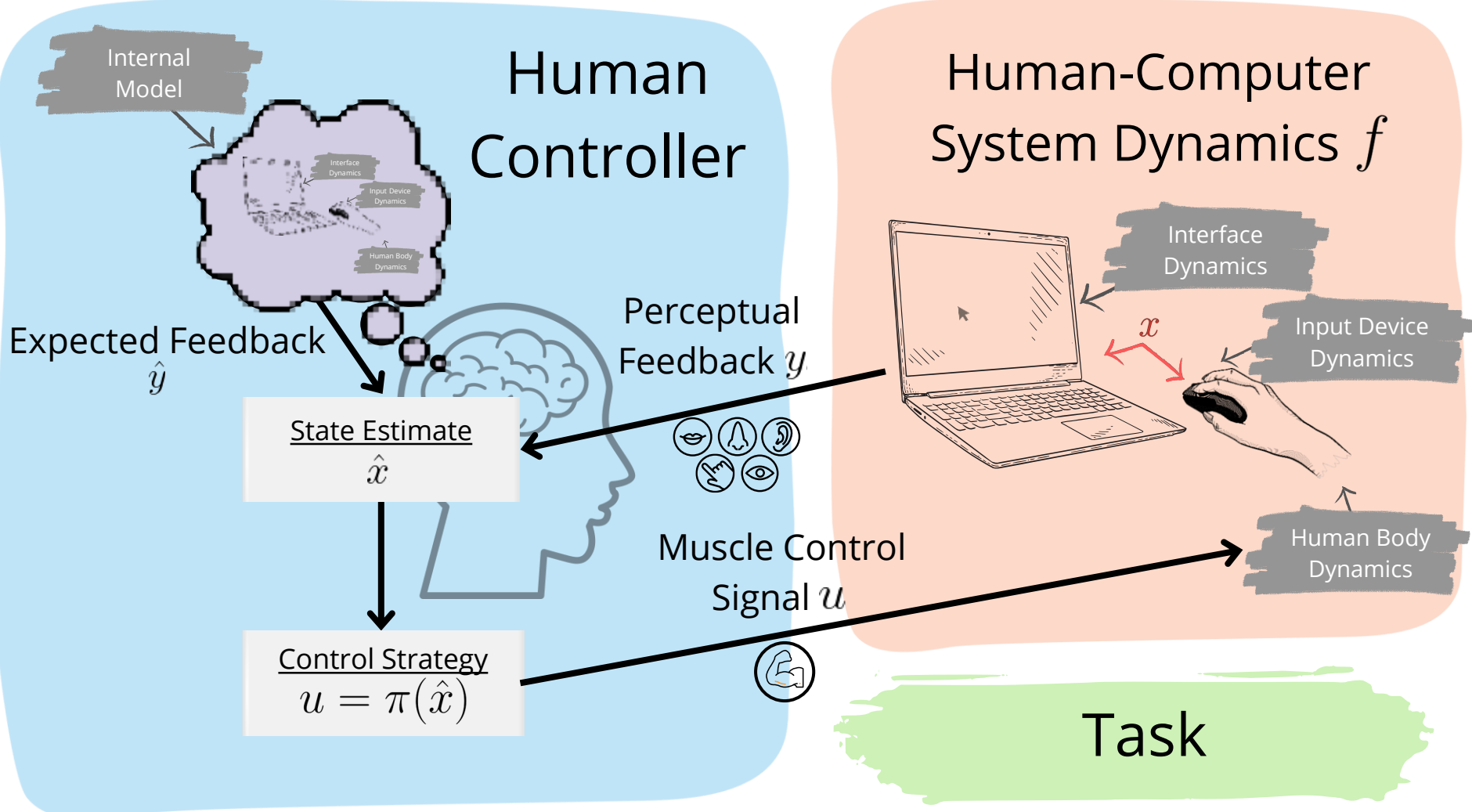
Muscle Control Signal u

Human Body Dynamics

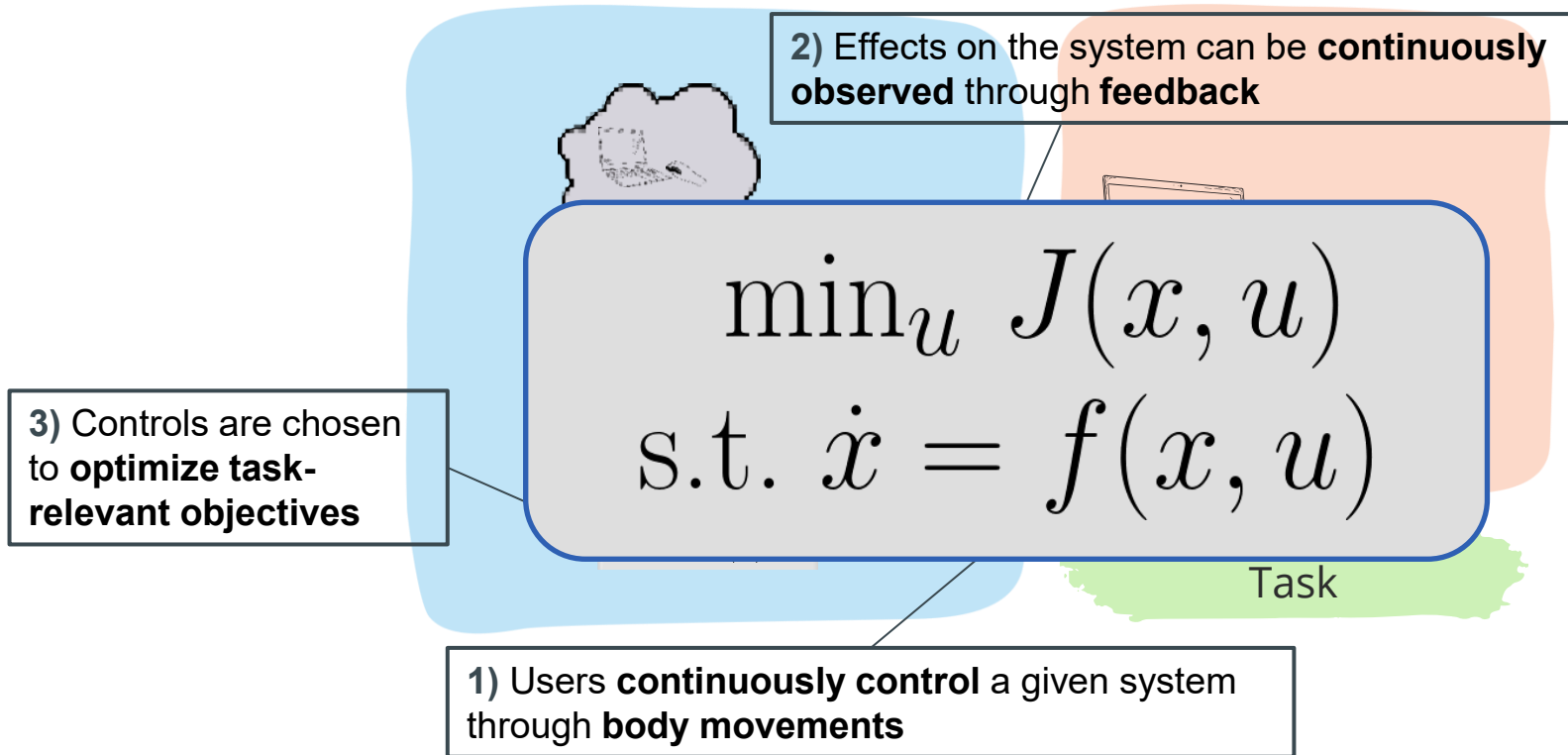
Costs J

Task





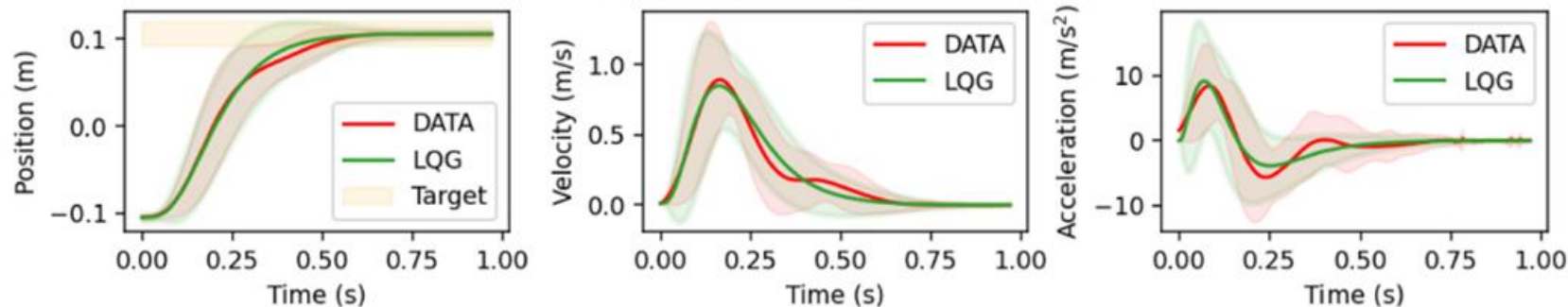
OFC for Movement-Based Interaction



Example: 1D Mouse Pointing (LQG)

$$J_N^{(\text{LQG})}(x, u) = \mathbb{E} \left[D_N^2 + \omega_v v_N^2 + \omega_f f_n^2 + \frac{\omega_r}{N-1} \left(\sum_{n=0}^{N-1} u_n^2 \right) \right]$$

LQG – P3, (765px, 51px), right direction

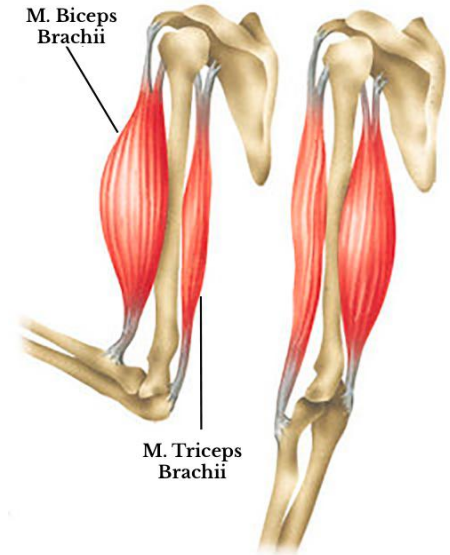


Muscle Redundancy Problem

- The human body has more muscles (~650) than DoFs (244)
- This implies there is an infinite number of ways to produce torque
- For each DoF, at least two muscles are available (antagonistic pair):

$$\tau = F_{m_a} \cdot r_{m_a} - F_{m_b} \cdot r_{m_b}$$

- Many muscles produce torques at several DoFs (primary moment, secondary moment, etc.), which requires further counterbalancing



3. Biomechanical RL for HCI

Biomechanical RL Procedure

- Define physical environment (“body model” and “interaction model”)
- Define task-specific reward (“user goals”)
- Define learning architecture (“brain model”)

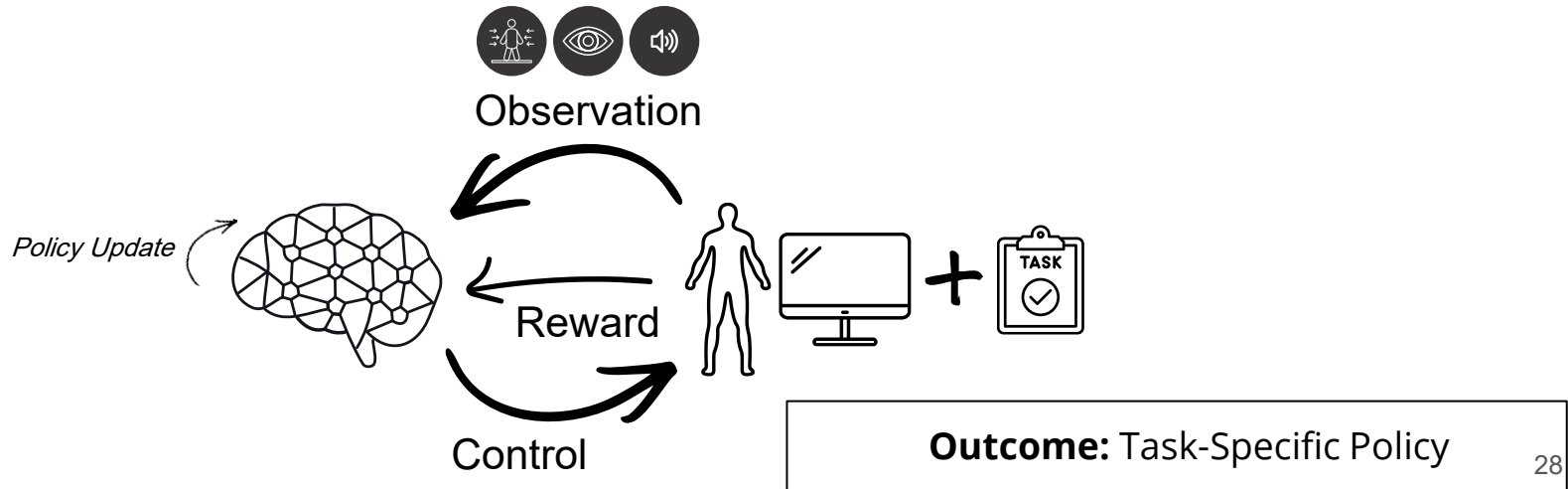


Biomechanical RL Procedure

Initialise policy (perceptuomotor control)

Iterate until convergence:

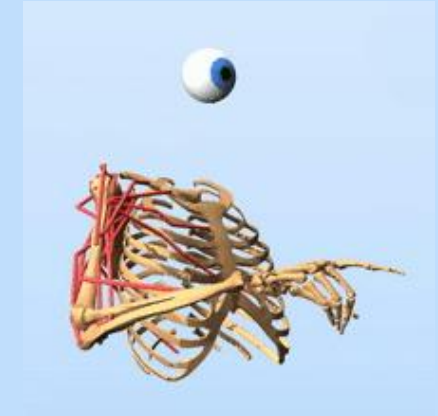
- Interact with the task, explore muscle controls
- Learn from obtained rewards



Biomechanical Simulations for HCI

- *MoBL ARMS* model of the upper extremity (MuJoCo)
- Fixed torso, wrist, and fingers (26 muscles)
- Visual and proprioceptive input

$$y_n = \begin{pmatrix} c_n \\ \omega_n \\ i_n \end{pmatrix} \begin{array}{l} \longleftarrow \text{RGB-D array} \in \mathbb{R}^{4 \times 80 \times 3} \\ \longleftarrow \text{Joint angles/vel./acc., muscle activation:} \\ \longleftarrow \text{Others} \in \mathbb{R}^2 \end{array}$$



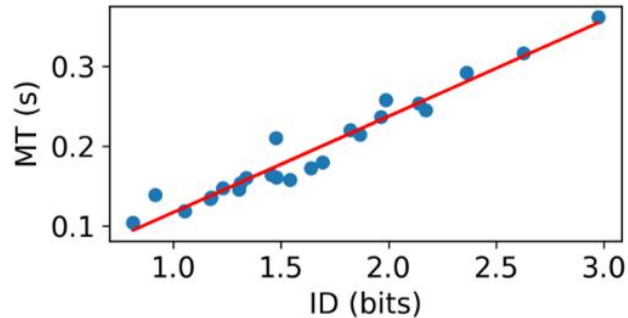
- Motor control policy is learned via Reinforcement Learning (PPO)

Predictions Follow Established Movement Laws

Pointing Task

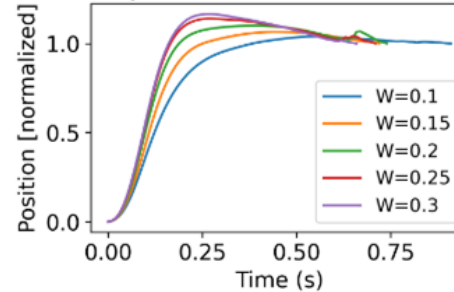
Fitts' Law

Fitts' Law on Mean Data
($R^2 = 0.9512$; $a = -0.003$, $b = 0.12$)

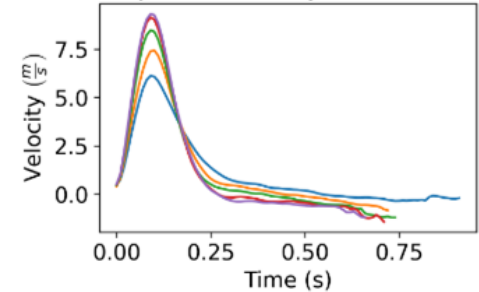


Effect of target size on end-effector movement

Projected Position Time Series



Projected Velocity Time Series



Ikkala, A., Fischer, F., Klar, M., Bachinski, M., Fleig, A., Howes, A., Hämmäläinen, P., Müller, J., Murray-Smith, R., & Oulasvirta, A. (2022). Breathing life into biomechanical user models. Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology. <https://doi.org/10.1145/3526113.3545689>

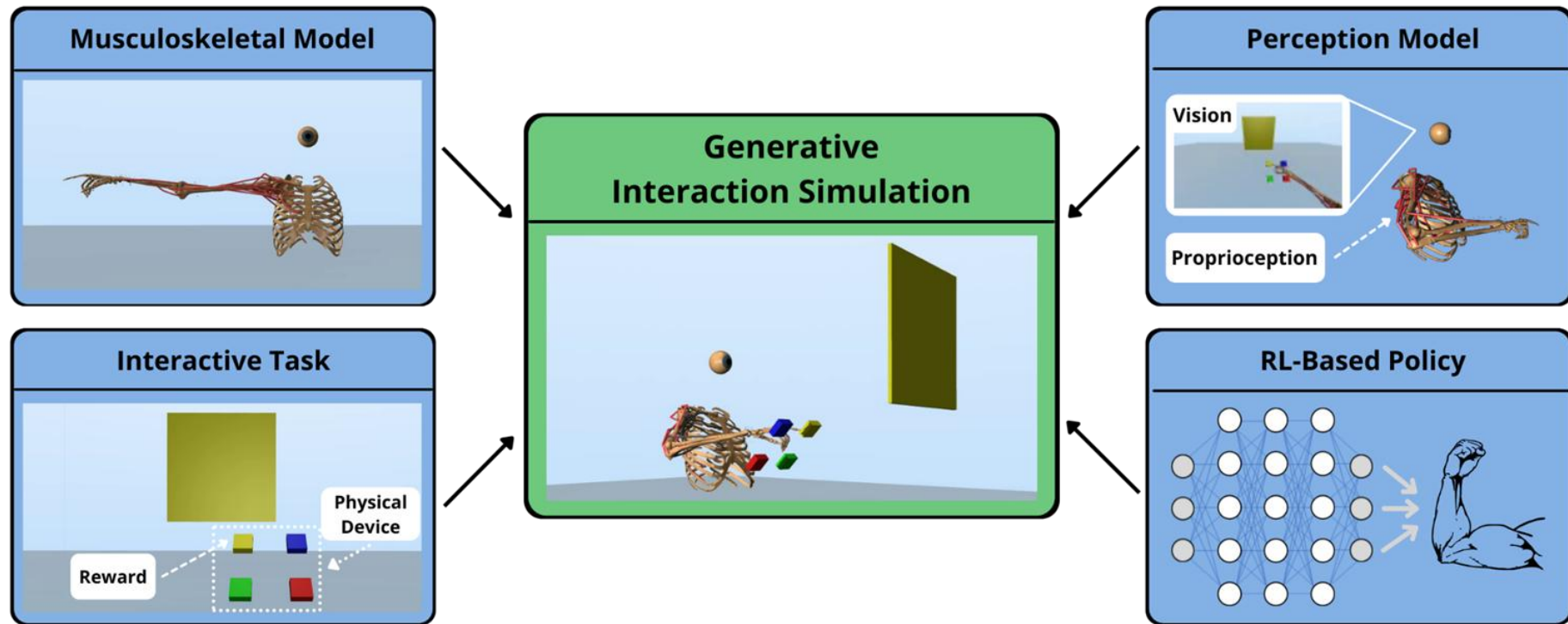
Examples: Visuomotor Interaction



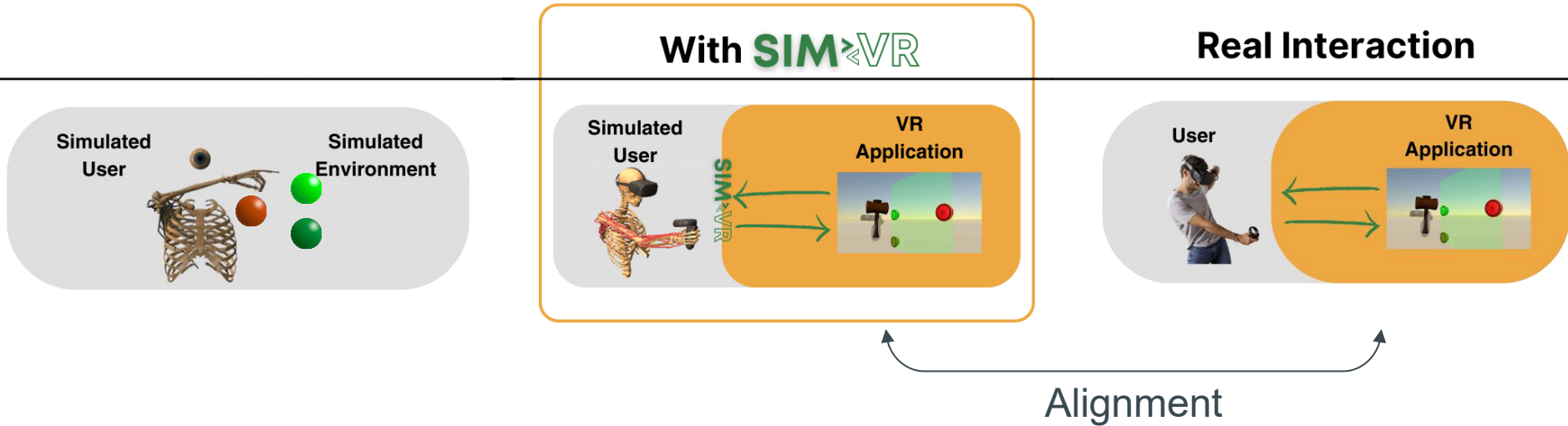
Ikkala, A., Fischer, F., Klar, M., Bachinski, M., Fleig, A., Howes, A., Hämäläinen, P., Müller, J., Murray-Smith, R., & Oulasvirta, A. (2022). **Breathing life into biomechanical user models.** Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology. <https://doi.org/10.1145/3526113.3545689>

User-in-the-Box

A Modular Simulation Framework

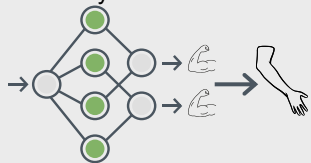


SIM $\rightleftharpoons$ VR

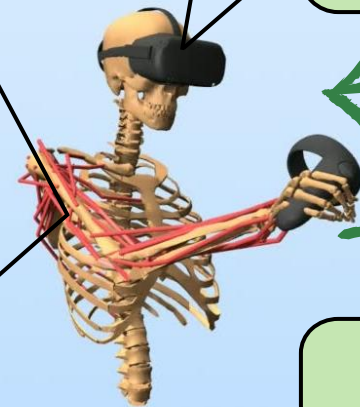


Simulated User

1) Select muscle control signals to generate body movement...



...and update position and orientation of the VR Controller and HMD

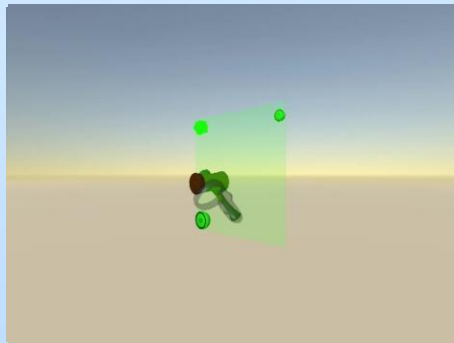


4) Send **rendered image** and **game reward** to the Simulated User

SIM <=> VR

2) Send **simulated sensor input** to the VR Application

VR Application



3) Update game physics and game logic...



...compute new reward signal...

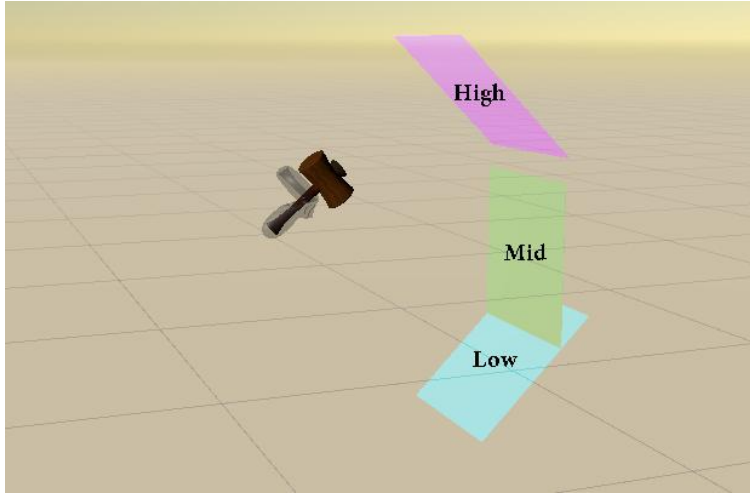


...and render the scene

Example: “Whac-A-Mole”



Height Conditions



Difficulty Conditions

- Easy (max. 1 target)
- Medium (max. 3 targets)
- Hard (max. 5 targets)

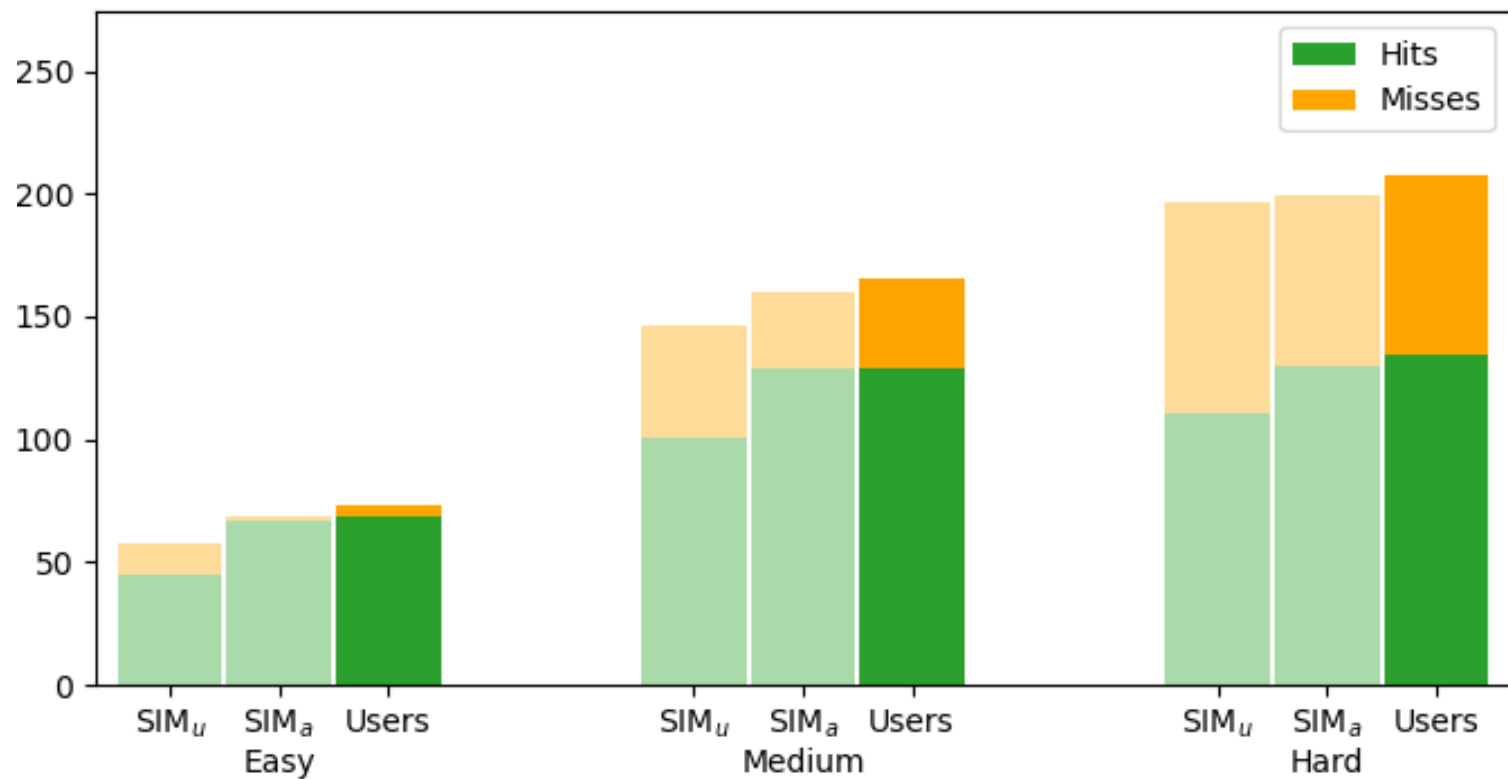
Simulated User



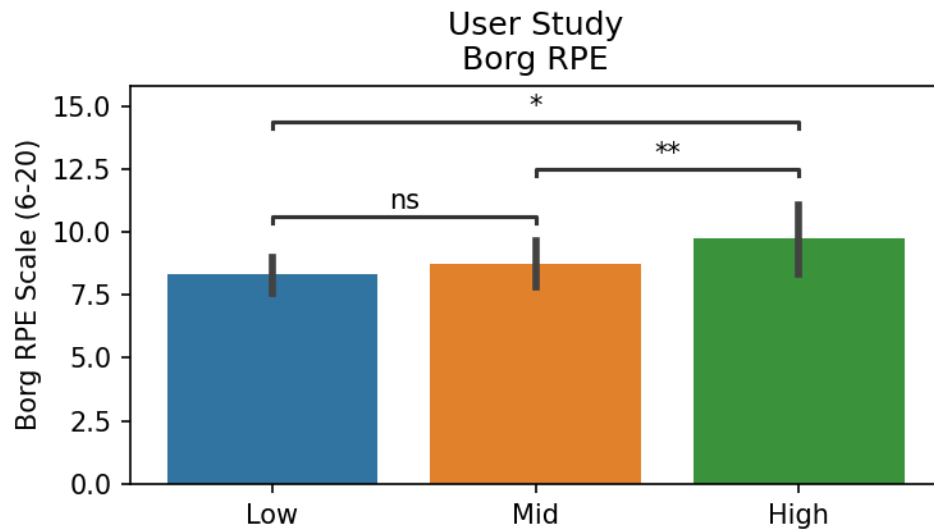
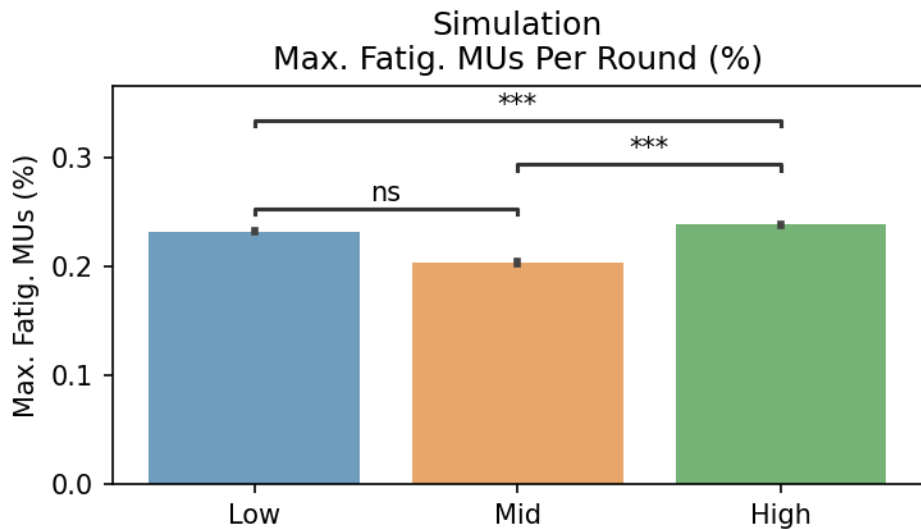
User Study (18 Participants)



Target Hits and Misses

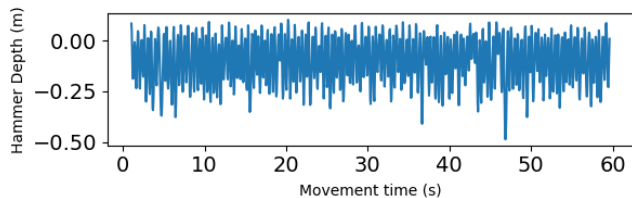
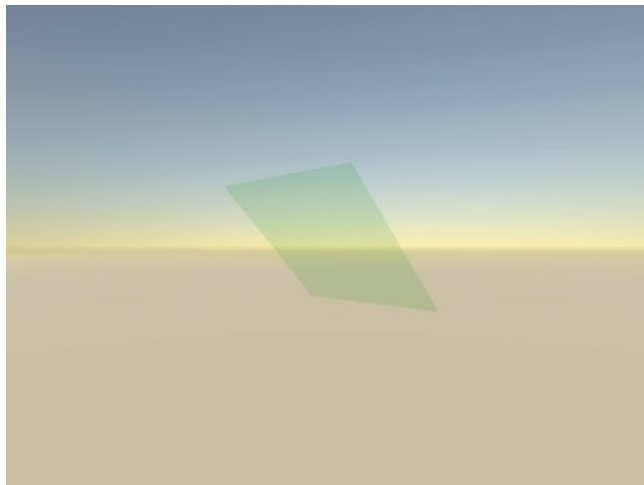


Effect of Height Condition on Effort

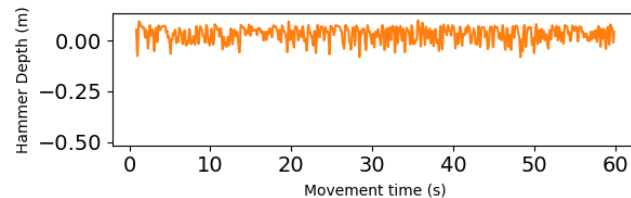
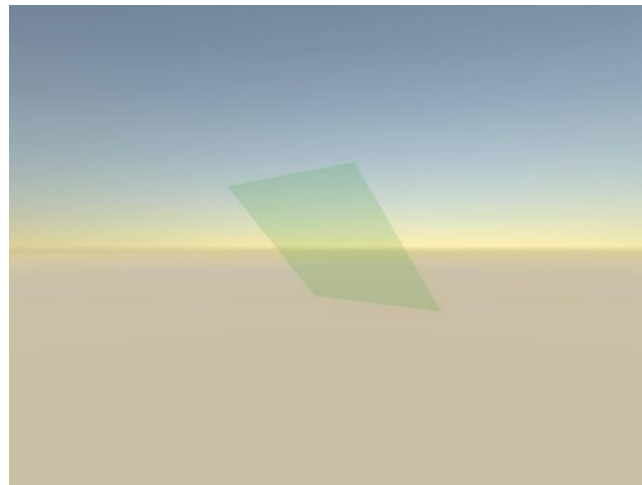


Simulated Users

Constrained

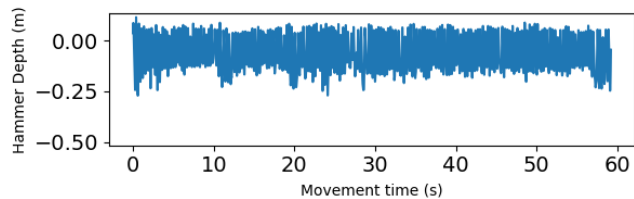
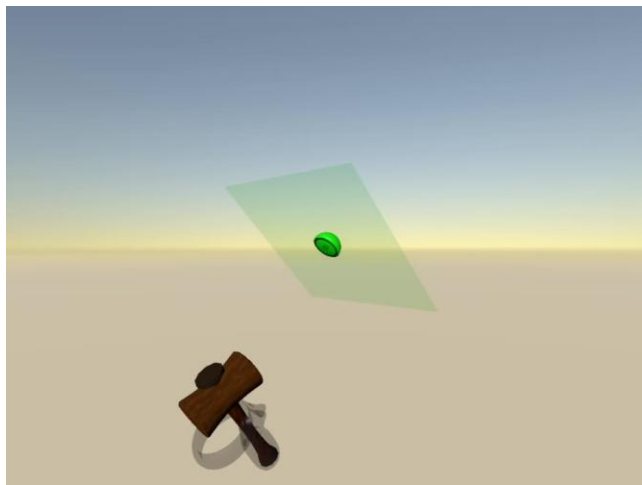


Unconstrained

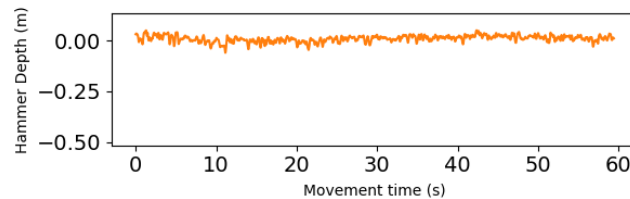
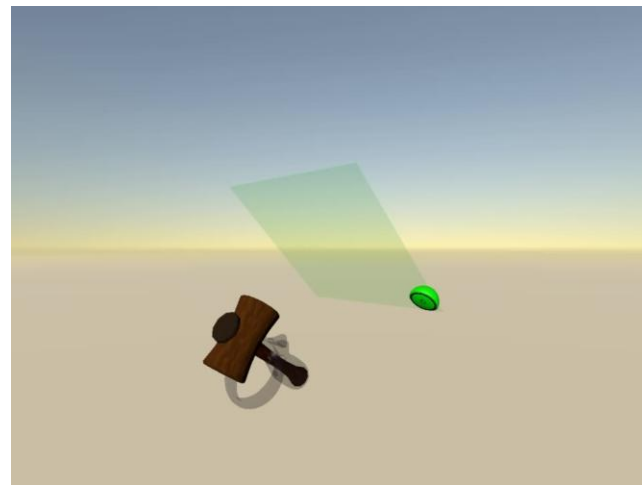


Real Users

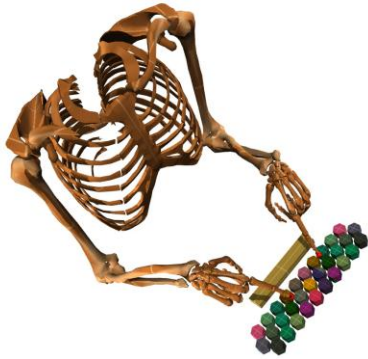
Constrained



Unconstrained



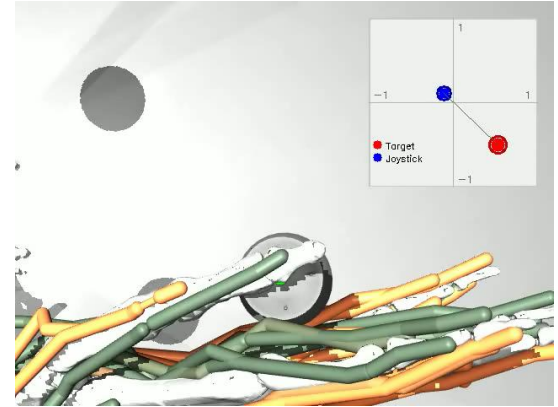
Simulating More HCI Tasks



Bimanual Keyboard Typing

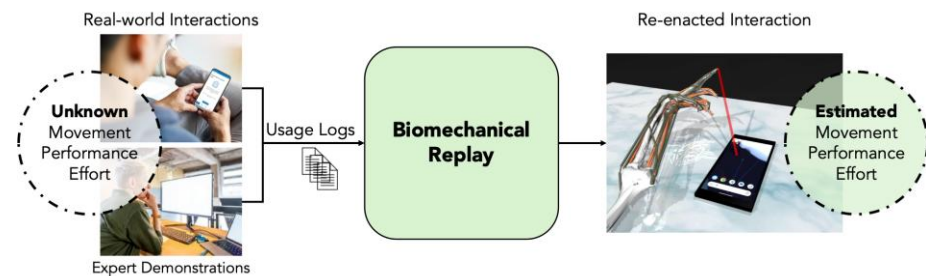


Dial Rotation

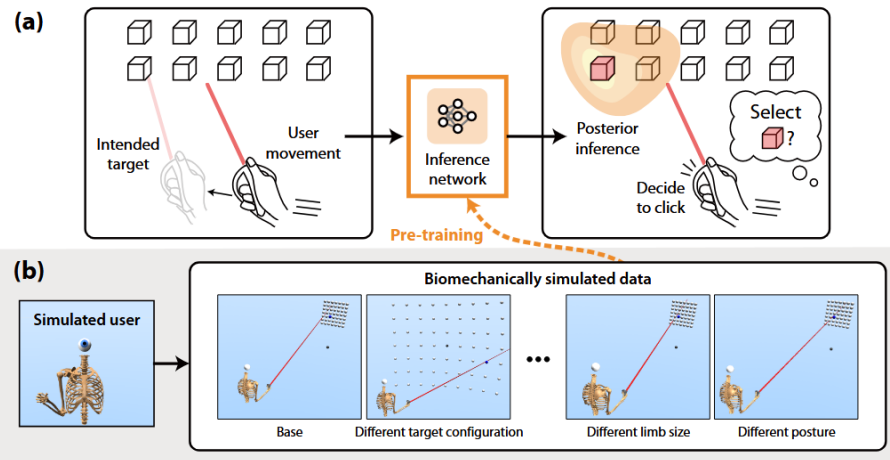


Joystick Input

Further Applications



Augment Interaction Traces
(*Log2Motion*,
Miazga et al. 2026)



Generate Synthetic Motion Datasets
(*Real-time 3D Target Inference*,
Moon et al. 2024)

4. Limitations and Bottlenecks

Challenge: Reward Function Design

- Reward function design:
a lot of **trial-and-error**
- +
- long training times (2-3 days for
one reward function)

→ **frustration**

- We performed a systematic
analysis of **>500 reward fcts.**

→ “Universal” **reward structure** +
step-by-step guidelines to (likely)
arrive at good agents

Training Reinforcement Learning
agents is a daunting task.

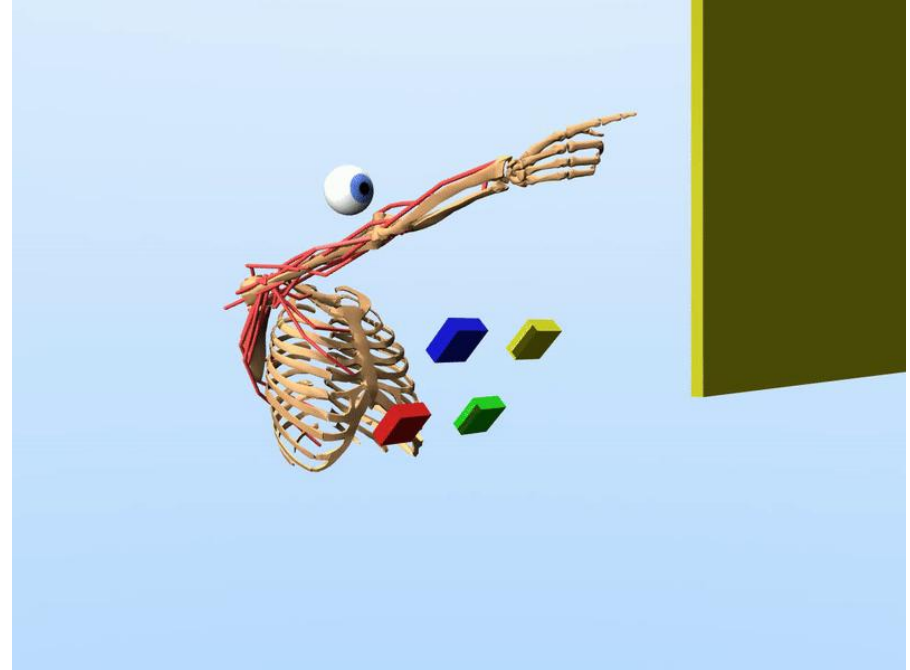
Challenge: Reward Function Design

Guidelines for r_t

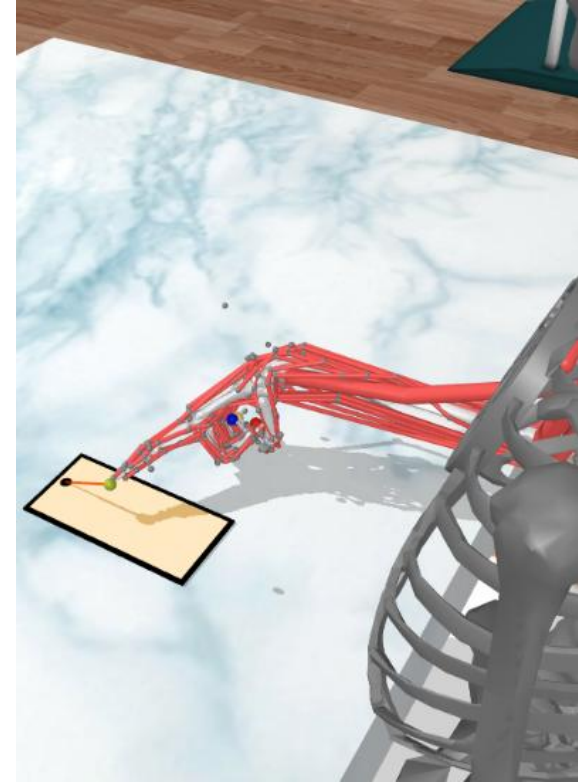
1. Use **guidance** components (e.g., distance term)
2. Include a **completion bonus**
3. Adapt the bonus **weight** (depends on one-time vs continuous bonus)
4. Try **without effort**; add effort term if movement instabilities occur
5. **Balance** distance and effort weights

Training Reinforcement Learning agents is a daunting task.

Simulated users often exhibit “weird”
and physiologically implausible
movements.



Simulated users might struggle with small targets or interactions that require high dexterity.



Learned strategies may not resemble
real user behaviour.



What could be the reason for undesired behaviour?

- Instability in the training
- Insufficient coverage of possible state-action pairs
- Inaccurate reward incentives
- Flaws/oversimplifications in the biomechanical model

Biomechanical RL is an iterative and interactive modelling process!

Limitations of Previous Biomechanical RL Frameworks

- **System Complexity → Hard to use**
- **Training Speed → Computationally extensive**
- **Measurability/Interpretability of Trained Policies → Hard to debug & assess**



MyoInteract

DIS²⁰₂₆

MyoInteract: A Framework for Fast Prototyping of Biomechanical HCI Tasks using Reinforcement Learning

Ankit Bhattarai*

Department of Engineering
University of Cambridge
Cambridge, United Kingdom
ab2731@cam.ac.uk

Hannah Selder*

Center for Scalable Data Analytics
and Artificial Intelligence (ScaDS.AI)
Dresden/Leipzig, Leipzig University
Leipzig, Germany
hannah.selder@uni-leipzig.de

Florian Fischer*

Department of Engineering
University of Cambridge
Cambridge, United Kingdom
fjf33@cam.ac.uk

Arthur Fleig[†]

Center for Scalable Data Analytics
and Artificial Intelligence (ScaDS.AI)
Dresden/Leipzig, Leipzig University
Leipzig, Germany
arthur.fleig@uni-leipzig.de

Per Ola Kristensson[†]

Department of Engineering
University of Cambridge
Cambridge, United Kingdom
pok21@cam.ac.uk

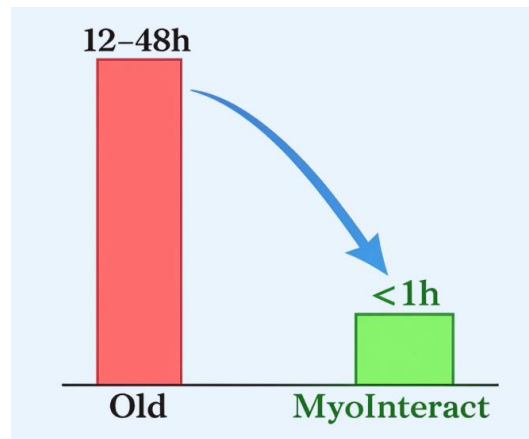
MyoInteract: A Framework for Fast Prototyping of Biomechanical HCI Tasks

Main Design Principles:

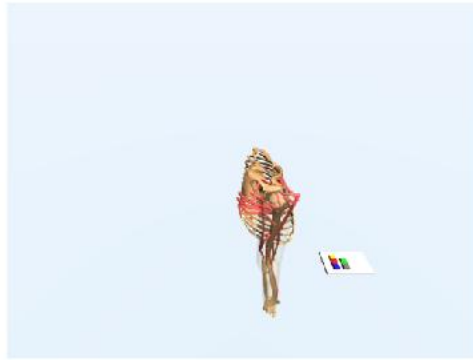
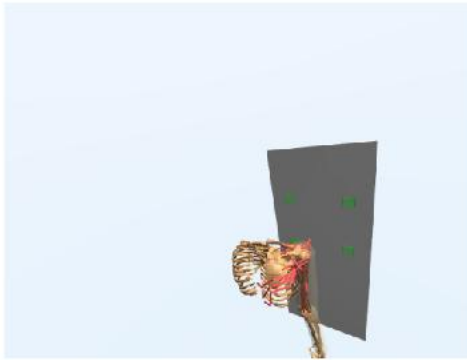
- **Fast user simulations through GPU-accelerated implementation of standard HCI tasks**
- **Generalisability & comparability through abstract task decomposition**
- **Multi-level feedback for diagnosis**
- **Encode domain constraints to prevent errors, “training wheels”**

MyoInteract: A Framework for Fast Prototyping of Biomechanical HCI Tasks

Experiment	Num Runs	Num Steps (M)	Total Training Time
UitB [12]	-	40-80	24-48 h
UitB (On Our Machine)	3	42.9 ± 9.0	6.9 ± 1.5 h
MyoInteract (Replication)	5	7.4 ± 0.8	10.3 ± 1.8 min
MyoInteract (Default)	5	34.9 ± 1.9	35.9 ± 2.0 min
MyoInteract + Warp (Replication)	5	8.8 ± 1.1	4.1 ± 0.5 min
MyoInteract + Warp (Default)	5	14.9 ± 4.4	6.9 ± 2.0 min



MyoInteract: An Interactive Tool for Biomechanical RL in HCI

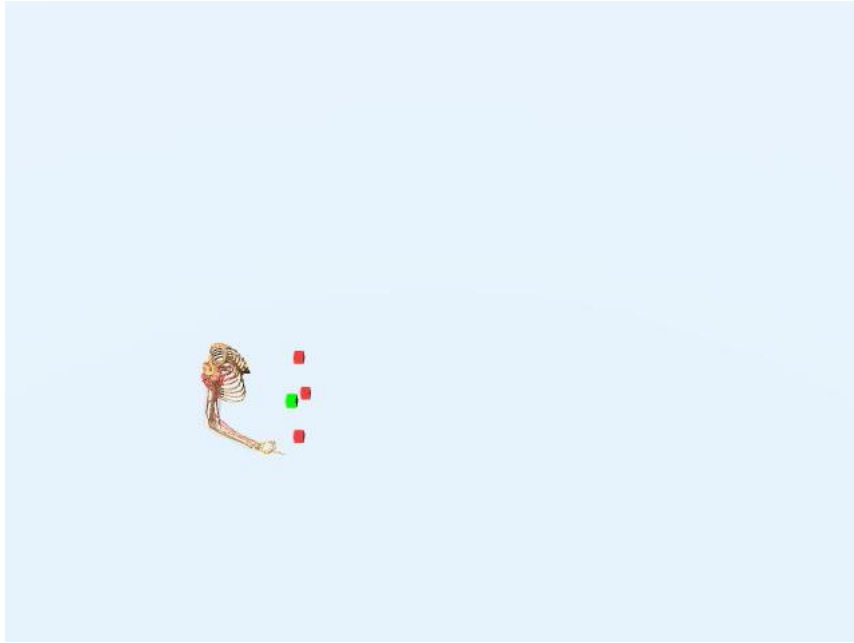


Demo: MyoInteract

→ bit.ly/s3cixbio



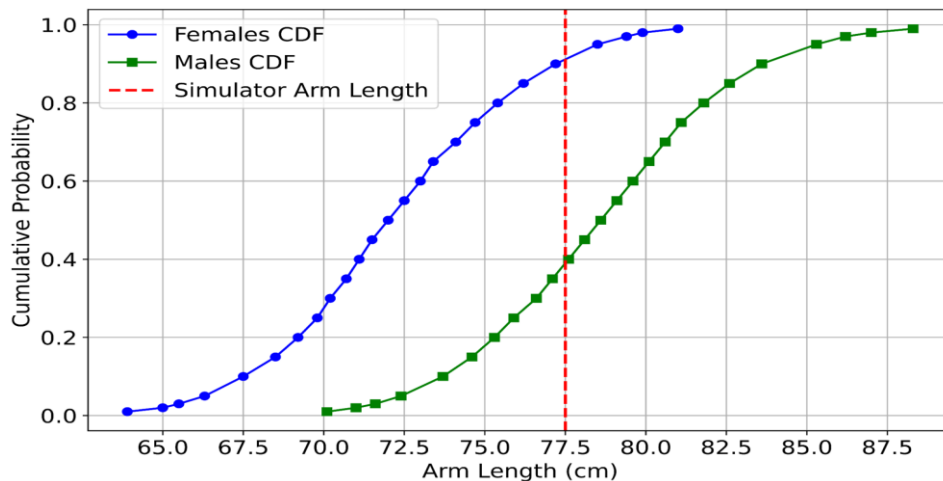
MyoInteract: An Interactive Tool for Biomechanical RL in HCI



What's next?

Open Challenges of Biomechanical User Simulations

- Simulating a target user population rather than an average user



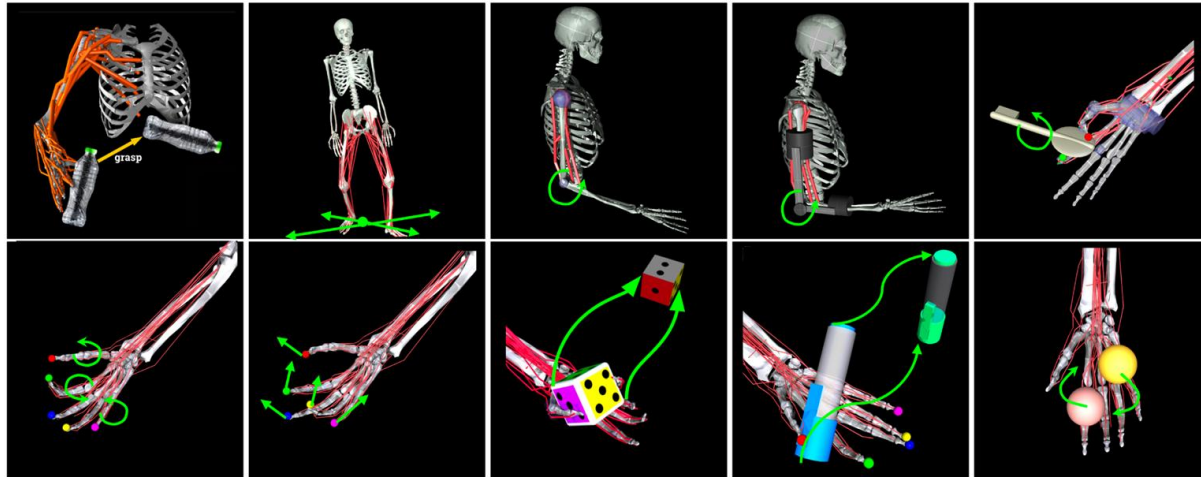
Miki, T., Fischer, F., Dudley, J.J. & Kristensson, P.O. (2025). Enhancing XR accessibility through anthropometrically diverse biomechanical simulation. *In Proceedings of the 1st IEEE VR 2025 Workshop on XR Accessibility (XRACCESS 2025)*.

Open Challenges of Biomechanical User Simulations

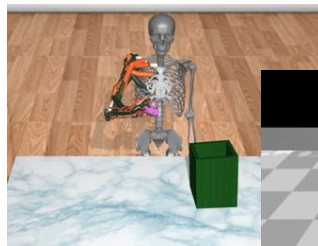
- HCI metrics and benchmarks to assess “human-likeness”
- Expanding the modelling scope to more realistic interfaces and interaction tasks
- Integrating perception models with biomechanical simulations
- Predicting longer time effects (injury risks, fatigue, muscle training)
- Extension to full-body models (with sufficient motor priors)
- ...

MyoSuite

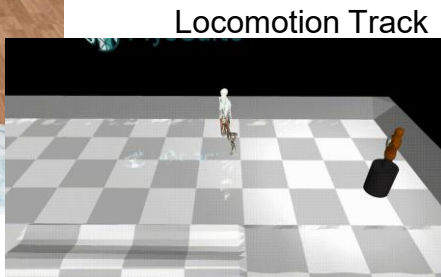
- Comprehensive set of 300+ tasks spanning upper and lower limbs to test human dexterity and agility



MyoChallenge @ NeurIPS



Manipulation Track



Locomotion Track



Learning Physiological
Dexterity

Towards Human-Level
Dexterity and Agility

Physiological Dexterity and Agility
in Enhanced Humans

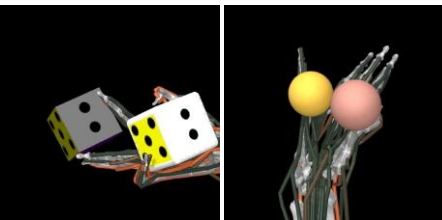
Towards Human Athletic
Intelligence

2022

2023

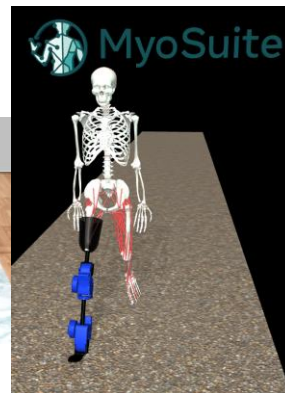
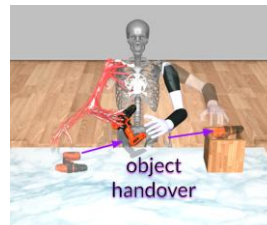
2024

2025



Die Rotation

Baoding Balls



Biomechanical Reinforcement Learning for HCI

Thank you for your participation!

Arthur Fleig, Florian Fischer, Miroslav Bachinski